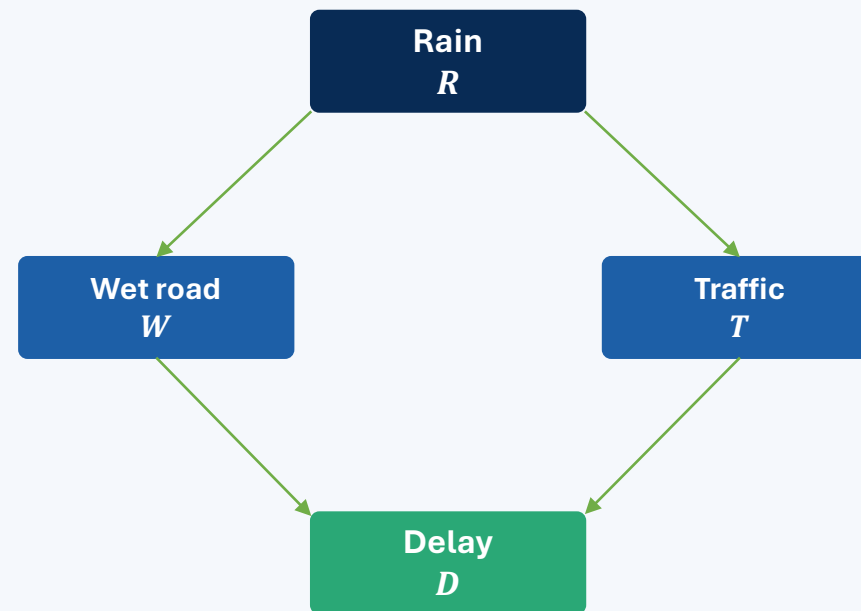


Artificial Intelligence

Lesson 21: Bayesian Inference

Query → Restrict → Sum Out → Normalize → Interpret



Ask a probability question of the model.

Inference turns a Bayesian network into a belief: what should we conclude from what we know?



Quiz 5 — 10 minutes

Use the first part of class to complete the probability, Bayes, and Bayes-net structure quiz.

Quiz 5

Probability foundations • Bayes' Rule • Bayesian-network structure

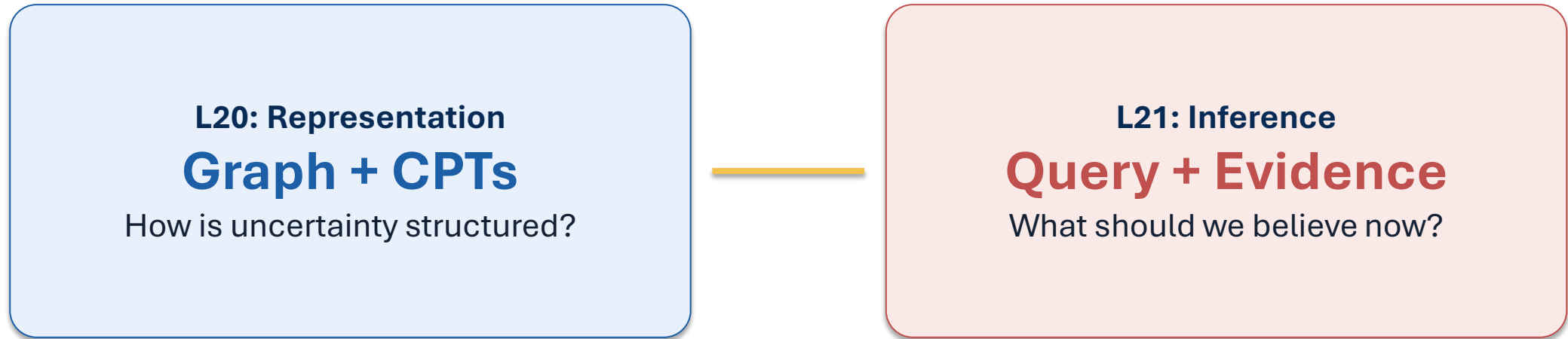
After the quiz:

Bayesian inference — using the network to answer probability queries.

When the quiz ends, we shift from representing a model to asking questions of the model.

From representation to inference

L20 built the model. L21 asks probability questions of it.



Same network. New task: compute and interpret probabilities from it.

Today's mission

A useful inference setup separates what we want, what we observed, and what remains unknown.

- 1 Identify query, evidence, and hidden variables.
- 2 Compute simple exact inference using factorization and marginalization.
- 3 Perform inference by enumeration on a small network.
- 4 Interpret posterior results and explain variable elimination conceptually.

Performance target: set up an inference problem before doing any arithmetic.

Three roles in an inference problem

Before computing, label what the probability question is asking.

Query variable

What are we trying to determine?

Evidence variable

What have we observed?

Hidden variable

What is neither queried nor observed?

Most inference errors start with confusing these roles.

You Try: label the variables

For the query below, decide which variables are query, evidence, and hidden.

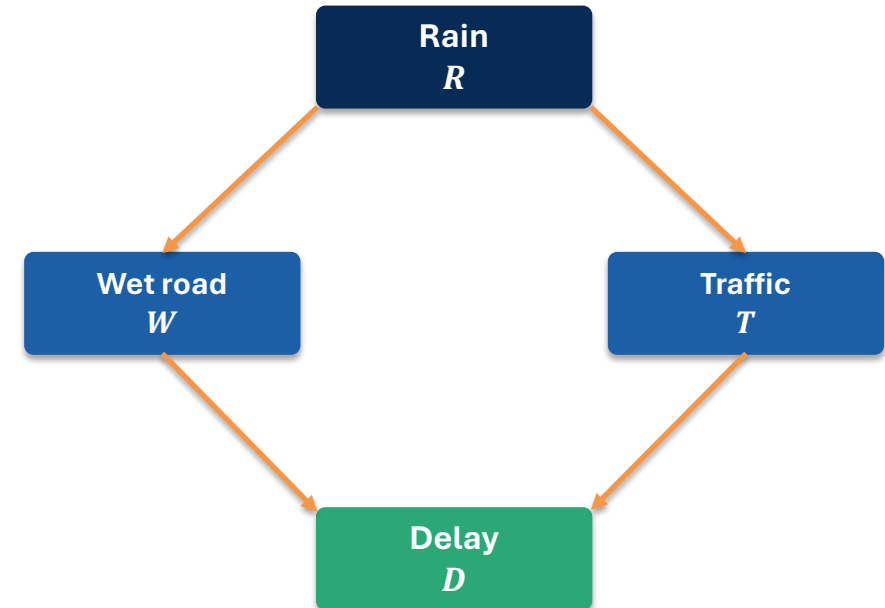
$$P(D \mid R = \text{true})$$

Decide with a partner

Query = ?

Evidence = ?

Hidden = ?



Answer: query D, evidence R, hidden W and T.

Inference is familiar probability reasoning

Start with one hidden variable before using the full network.

$$P(M) = P(M | V)P(V) + P(M | \neg V)P(\neg V)$$

What are we doing?

Weighting each possible case of V ,
then summing.

Why?

V is hidden:
not queried and not observed.

Inference often means summing over possible hidden cases.

Inference is marginalization

Use the L18 idea: keep what matters and sum out what you do not need.

$$\begin{aligned} P(M) &= \sum_v P(M, v) \\ &= \sum_v P(M | v) P(v) \end{aligned}$$

Plain English

For each possible hidden value, compute its contribution,
then add the contributions together.

This same sum-over-hidden-cases idea will return in Markov models and HMMs.

You Try: one hidden variable

Compute a probability by summing over the possible values of the hidden variable.

Given

$$P(S) = 0.60$$

$$P(A | S) = 0.80$$

$$P(A | \neg S) = 0.20$$

Find

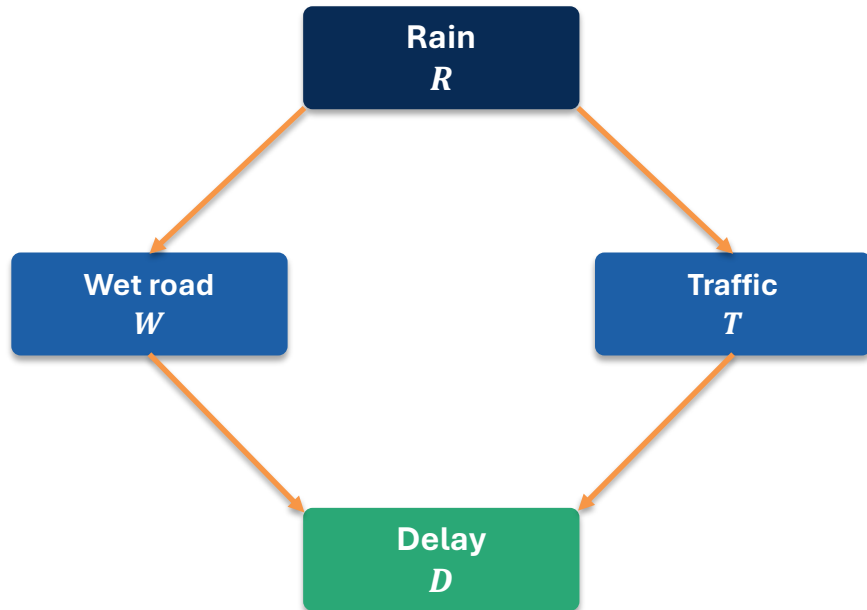
$$P(A) = ?$$

Use both hidden cases: S and $\neg S$.

$$\text{Solution: } P(A) = (0.80)(0.60) + (0.20)(0.40) = 0.56$$

Factorization gives us the pieces

The Bayesian network tells us which local probability tables to multiply.



L20 factorization

$$P(R, W, T, D) = P(R)P(W|R)P(T|R)P(D|W, T)$$

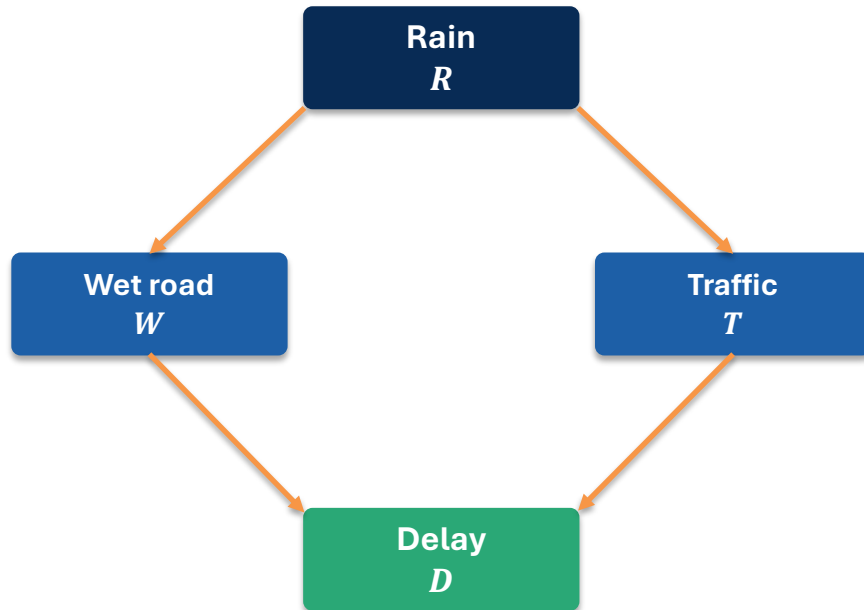
Which variables matter for delay once rain is known?

For $P(D | R)$: use $P(W|R)$, $P(T|R)$, and $P(D|W, T)$.

The graph structure tells us which local tables participate in the inference.

Query: $P(D \mid R)$

Evidence is known. Hidden variables still carry uncertainty.



$$P(D \mid R) = \sum_{w,t} P(w \mid R) P(t \mid R) P(D \mid w, t)$$

Interpretation before arithmetic

Consider every possible combination of wet road and traffic, then add their contributions to delivery delay.

Hidden variables are not ignored. They are summed out.

Enumeration means considering all hidden assignments

For binary W and T , there are four hidden cases.

$$P(D | R) = \sum_{w,t} P(w | R) P(t | R) P(D | w, t)$$

Case	W	T	Term
1	true	true	$P(w R)P(t R)P(D w, t)$
2	true	false	$P(w R)P(\neg t R)P(D w, \neg t)$
3	false	true	$P(\neg w R)P(t R)P(D \neg w, t)$
4	false	false	$P(\neg w R)P(\neg t R)P(D \neg w, \neg t)$

Enumeration is systematic: list cases, compute each contribution, then sum.

Worked enumeration table

Use simple local probabilities to compute $P(D | R)$.

W	T	$P(W R)$	$P(T R)$	$P(D W, T)$	Contribution
true	true	0.80	0.60	0.90	0.432
true	false	0.80	0.40	0.60	0.192
false	true	0.20	0.60	0.70	0.084
false	false	0.20	0.40	0.10	0.008

$$P(D | R) = \sum_{w,t} P(w | R) P(t | R) P(D | w, t)$$

$$P(D | R) = 0.432 + 0.192 + 0.084 + 0.008 = 0.716$$

Compute → Interpret → Explain

A numerical answer is not enough; students must explain the inference.

Compute

$$P(D | R) = 0.716$$

Interpret

**Given rain,
the model assigns 71.6%
probability to delayed
delivery.**

Explain

**Wet road and traffic are
hidden pathways, so we
summed over their
possible values.**

Inference results should always be interpreted as beliefs under the model, not guarantees.

When do we normalize?

If we compute unnormalized query scores, divide by their total to make a distribution.

$$P(D \mid e) = \frac{P(D, e)}{P(D, e) + P(\neg D, e)}$$

Unnormalized scores

D : 0.18

$\neg D$: 0.42

Normalize

total = 0.60

$P(D|e) = 0.18/0.60 = 0.30$

Select evidence → sum hidden variables → normalize the query distribution.

You Try: normalize

Turn unnormalized query scores into a probability distribution.

Unnormalized

$Q = \text{true}: 0.24$

$Q = \text{false}: 0.56$

Find

$P(Q = \text{true} \mid e)$

$P(Q = \text{false} \mid e)$

Solution: total = 0.80,
so $P(Q = \text{true} \mid e) = 0.30$ and $P(Q = \text{false} \mid e) = 0.70$.

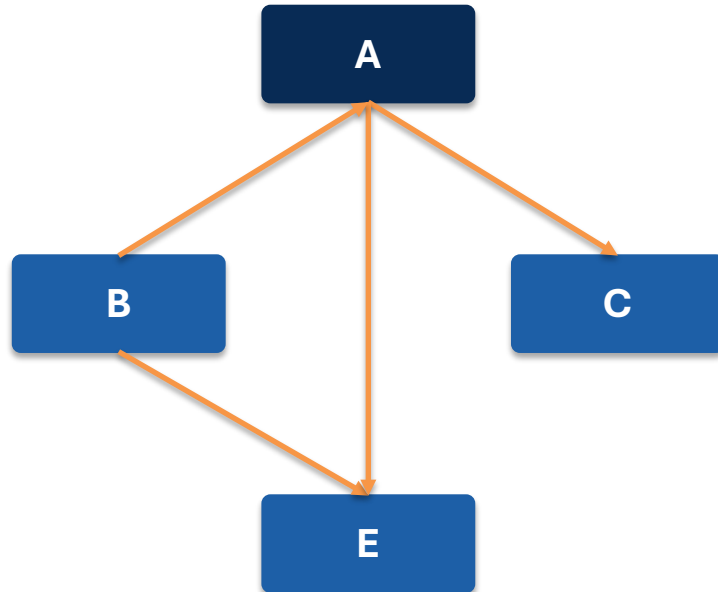
Inference-by-enumeration workflow

A reliable setup prevents most mistakes.

- 1 Identify query and evidence.
- 2 Identify hidden variables.
- 3 Use the Bayes-net factorization.
- 4 Keep assignments consistent with evidence.
- 5 Sum over hidden variables.
- 6 Normalize if computing a conditional distribution.
- 7 Interpret the result.

You Try: plan the inference

Do the setup before doing any arithmetic.



Scenario

- **B**: High server load
- **A**: Resource contention
- **C**: High latency
- **E**: request failure

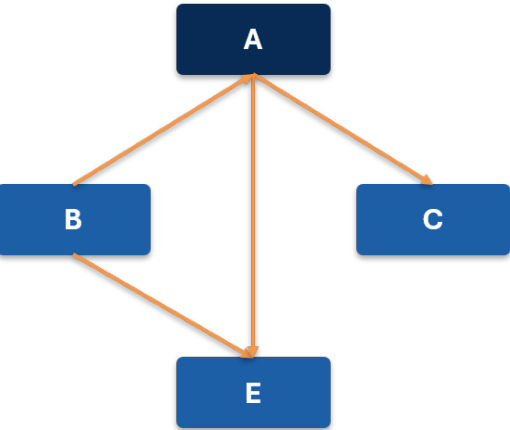
$$P(A, B, C, E) = P(B)P(A|B)P(C|A)P(E|A, B)$$

If a request did not fail, what is the probability the service was experiencing high latency?

What is the query, evidence, and hidden variables? How many hidden assignments are there?

You Try: calculate the inference

Now let's do the arithmetic.



Scenario

- **B:** High server load
- **A:** Resource contention
- **C:** High latency
- **E:** request failure

$$P(A, B, C, E) = P(B)P(A|B)P(C|A)P(E|A, B)$$

If a request did not fail, what is the probability the service was experiencing high latency?

B	P(B)
true	0.40
false	0.60

A	B	P(A B)
true	true	0.80
true	false	0.80

C	A	P(C A)
true	true	0.90
true	false	0.20

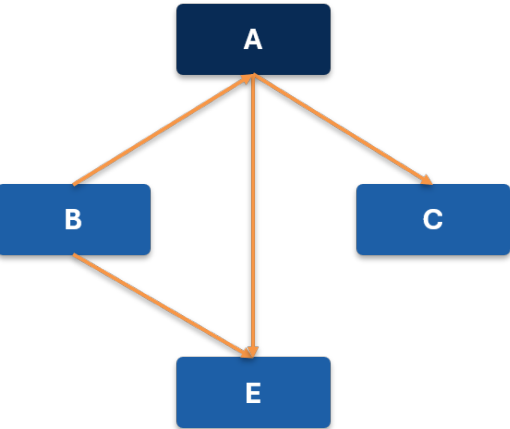
E	A	B	P(E A,B)
true	true	true	0.90
true	true	false	0.60
true	false	true	0.40
true	false	false	0.10

$$P(C|E = \text{false}) = \frac{P(C, \neg E)}{P(\neg E)} = \frac{\sum_{a,b} P(b)P(a|b)P(C|a)P(\neg E|a, b)}{\sum_{a,b} P(b)P(a|b)P(\neg E|a, b)}$$

B	A	P(B)	P(A B)	P(C A)	P(E A,B)	P(¬E A,B)	P(C ¬E)	P(¬E)
T	T	.40	.80	.90	.90	.10	.0288	.032
T	F	.40	.20	.20	.40	.60	.0096	.048
F	T	.60	.20	.90	.60	.40	.0432	.048
F	F	.60	.80	.20	.10	.90	.0864	.432
						Sum	.168	.560

You Try: calculate the inference

Now let's do the arithmetic.



Scenario

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$$P(A, B, C, E) = P(B)P(A|B)P(C|A)P(E|A, B)$$

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true	0.40
false	0.60

A	B	P(A B)
true	true	0.80
true	false	0.80

C	A	P(C A)
true	true	0.90
true	false	0.20

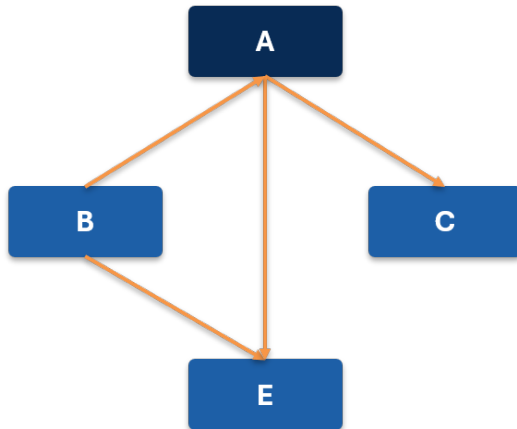
E	A	B	P(E A,B)
true	true	true	0.90
true	true	false	0.60
true	false	true	0.40
true	false	false	0.10

$$P(C|E = \text{false}) = \frac{P(C, \neg E)}{P(\neg E)} = \frac{\sum_{a,b} P(b)P(a|b)P(C|a)P(\neg E|a, b)}{\sum_{a,b} P(b)P(a|b)P(\neg E|a, b)}$$

B	A	P(B)	P(A B)	P(C A)	P(E A,B)	P(¬E A,B)	P(C ¬E)	P(¬E)
T	T	.40	.80	.90	.90	.10	.0288	.032
T	F	.40	.20	.20	.40	.60	.0096	.048
F	T	.60	.20	.90	.60	.40	.0432	.048
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You Try: calculate the inference

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Scenario

- **B:** High server load
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$$P(A, B, C, E) = P(B)P(A|B)P(C|A)P(E|A, B)$$

If a request did not fail, what is the probability the service was experiencing high latency?

B	P(B)
true	0.40
false	0.60

A	B	P(A B)
true	true	0.80
true	false	0.80

C	A	P(C A)
true	true	0.90
true	false	0.20

E	A	B	P(E A,B)
true	true	true	0.90
true	true	false	0.60
true	false	true	0.40
true	false	false	0.10

$$P(C|E = \text{false}) = \frac{P(C, \neg E)}{P(\neg E)} = \frac{\sum_{a,b} P(b)P(a|b)P(C|a)P(\neg E|a, b)}{P(\neg E)}$$

$$P(C, \neg E) = (0.4)(0.8)(0.9)(0.1) + (0.4)(0.2)(0.2)(0.6) + (0.6)(0.2)(0.9)(0.4) + (0.6)(0.8)(0.2)(0.9) \\ = 0.0288 + 0.0096 + 0.0432 + 0.0864 = 0.168$$

$$P(\neg E) = (0.4)(0.8)(0.1) + (0.4)(0.2)(0.6) + (0.6)(0.2)(0.4) + (0.6)(0.8)(0.9) \\ = 0.032 + 0.048 + 0.048 + 0.432 = 0.56$$

$$P(C|\neg E) = 0.168 / 0.56 = 0.30$$

Enumeration works—but it can grow quickly

The method is exact, but hidden assignments multiply.

5 hidden binary variables

$$2^5 = 32$$

assignments

10 hidden binary variables

$$2^{10} = 1024$$

assignments

Problem

Enumeration may repeat intermediate calculations many times.

This is why more advanced inference algorithms exist.

Variable elimination: same math, smarter order

The idea is to interleave multiplication and marginalization.

Enumeration

Build many hidden assignments, then sum out hidden variables.

Variable elimination

Combine only the factors mentioning a hidden variable, sum it out, then continue.

For CS 471, master small enumeration; recognize variable elimination as the efficient exact-inference idea.

Tiny variable-elimination illustration

Same answer, less repeated computation.



Enumeration: $P(C) = \sum_{a,b} P(a)P(b|a)P(C|b)$

VE Step 1: $f(b) = \sum_a P(a)P(b|a)$

VE Step 2: $P(C) = \sum_b f(b)P(C|b)$

Key idea

Create smaller intermediate summaries instead of repeatedly rebuilding the same pieces.

What to master vs. recognize

This keeps the lesson rigorous without overloading the first exposure.

Master in CS 471

- query / evidence / hidden roles
- Bayes-net factorization
- marginalization
- small enumeration
- normalization
- interpretation

Recognize for advanced AI

- factors
- variable elimination
- exact inference complexity
- approximate inference / sampling

The conceptual bridge matters even when the full algorithm is beyond this lesson.

Representation vs. inference

The last two lessons are two halves of the same modeling story.

L20: Representation **Graph + CPTs**

How is uncertainty structured?

L21: Inference

Factorize + Sum Out + Normalize

What should we believe given evidence?

The representation makes inference possible; inference is why we built the representation.

The recurring probability pattern

The same weighted-sum idea keeps reappearing across Block 3.

$$\sum_x P(y | x) P(x)$$

L18

marginalization

L19

evidence term

L21

Bayes-net inference

L22

state propagation

L23

HMM prediction

We are not learning a brand-new equation each lesson; we are reusing the same inference idea.

Exit ticket

Complete before you leave.

Fill in the blanks

Query variable = _____

Evidence variable = _____

Hidden variable = _____

Inference often handles hidden variables by _____ them out.

One inference result from today means _____.

Next: what changes when our uncertain variable evolves through time?